East West University

Department of Computer Science & Engineering

Task 1 Report

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Course Name:	Machine Learning
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Title:	Exploratory Data Analysis and Related-Work Review for Four-Class Brain Tumor Classification from MRI Images
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Abstract

This report presents an exploratory data analysis (EDA) of the Brain Tumor MRI Dataset (Kaggle copy by M. Nickparvar) and a review of five related deep-learning studies on brain tumor detection from MRI. The dataset contains 7,200 JPEG images in four classes (glioma, meningioma, pituitary tumor and no tumor), split into 5,600 training and 1,600 testing images. The analysis covers folder structure, class balance, image size and color mode, visual inspection, pixel statistics, corrupted files, exact duplicates, train/test leakage, blur and brightness. The classes are perfectly balanced in the original release, but image sizes are very heterogeneous (447 unique sizes) and 187 exact duplicate files were found (171 in Training, 16 in Testing), none shared between the two splits. After duplicate removal the Training set has 5,429 images and the class ratio becomes about 1.09. Since this task does not involve training a classifier, no model performance is claimed.

1. Introduction and Objectives

Magnetic resonance imaging (MRI) is a standard tool for finding and characterizing brain tumors, and convolutional neural networks (CNNs) are widely used to classify tumor type automatically. Before any model is designed, the data must be understood: how many images each class has, how images differ in size and color mode, whether any files are broken or repeated, and whether the test set is truly independent of the training set. Errors in these areas can inflate reported accuracy without improving the model.

The objectives of Task 1 are:

- to build a reproducible EDA baseline for the Brain Tumor MRI Dataset;
- to check data quality (corrupted files, duplicates, leakage, blur and brightness) and decide a cleaning rule;
- to derive preprocessing decisions (input size, normalization values, class weights, split policy);
- to review five related studies and identify gaps that our later experiments can address.

2. Dataset and Analysis Method

The notebook was run on Kaggle. It detects the folder that contains Training and Testing, builds a DataFrame with the path, split, class, file extension and file size of every image, and then adds width, height and color mode. A fixed random seed (42) is used for all sampling so the results can be reproduced. Counts, formats, sizes, corruption checks and hashing cover all 7,200 images. Pixel statistics and the class mean images use random samples of Training images (1,000 images, grayscale, resized to 128×128, and 200 images per class). The blur and brightness checks use a random sample of 1,500 images from both splits.

A. Dataset Summary

Property	Observed description
Name / source	Brain Tumor MRI Dataset, Kaggle (masoudnickparvar/brain-tumor-mri-dataset)
Application area	Multi-class brain tumor classification from MRI
Total images	7,200 (all .jpg)
Splits	Training 5,600; Testing 1,600
Classes	4: glioma, meningioma, notumor, pituitary
Images per class	Training 1,400 each; Testing 400 each
Color modes	RGB 4,129; L (grayscale) 3,067; RGBA 3; P 1
Image sizes	447 unique (width, height) pairs; 512×512 is the most common (5,014 images)
File size	Mean 22.77 KB, median 21.20 KB, min 3.59 KB, max 299.31 KB
Corrupted files	0
Exact duplicates	187 duplicate files (171 in Training, 16 in Testing); 340 images involved in a duplicate group
Train/Test overlap	0 images shared between Training and Testing

The acquisition details of the images (scanner, MRI sequence, patient information and preprocessing history) are not provided in the folder structure and were not independently verified.

B. Class Balance

The original release is perfectly balanced: every class has 1,400 Training images and 400 Testing images, so the max/min imbalance ratio in Training is 1.00 and the suggested class weights are all 1.0 (Figure 1). Class weighting is therefore not needed for the original data. After duplicate removal (Section 2F) the Training counts change slightly.

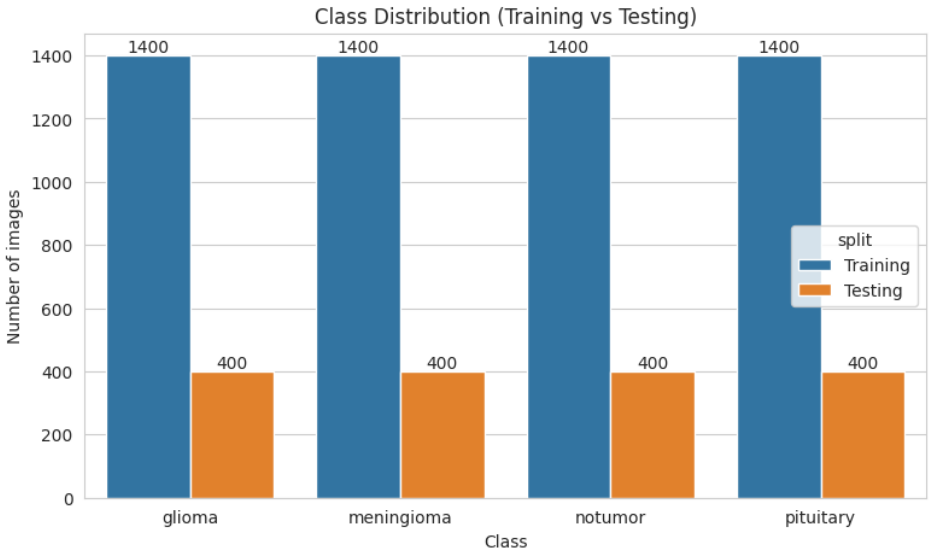


Figure 1. Class distribution of the Training and Testing splits (original data).

C. Image Size and Color Mode

Images are not uniform (Figure 2). About 69.6% (5,014 of 7,200) are 512×512, while the remaining 2,186 images are spread over 446 other sizes, including non-square ones such as 201×251 and 150×198. Color modes are also mixed: 3,067 images are single-channel grayscale (L), 4,129 are RGB, and 4 images are RGBA or palette (P). Every image must therefore be resized to a fixed size and converted to one consistent channel format before training. Mixed sizes and modes also suggest that the images may come from different scanners or sources, which could affect generalization.

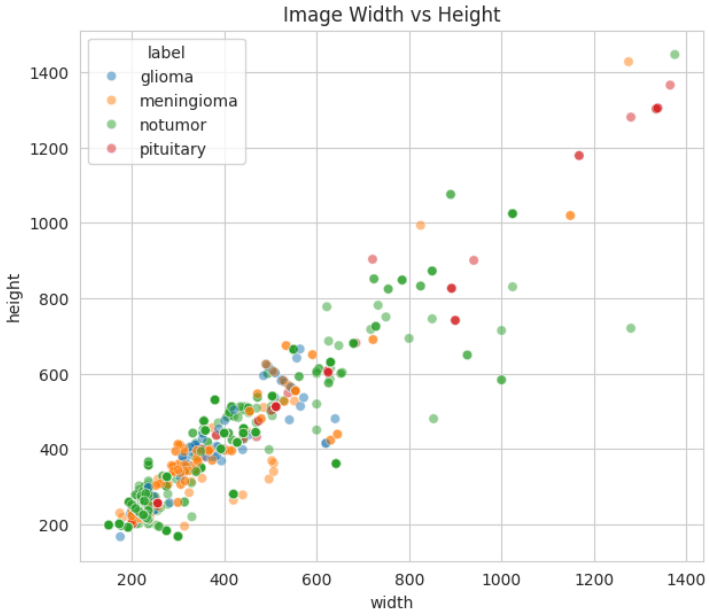


Figure 2. Image width versus height, colored by class.

D. Visual Inspection

Five random Training images per class were displayed (Figure 3). The sample contains axial, sagittal and coronal views, and the image contrast and intensity appearance differ visibly between samples, including within the same class. One sampled no-tumor image carries a burned-in number in its corner, which shows that some images contain annotations or other artifacts. This variation in view and appearance is a point to keep in mind for later training.

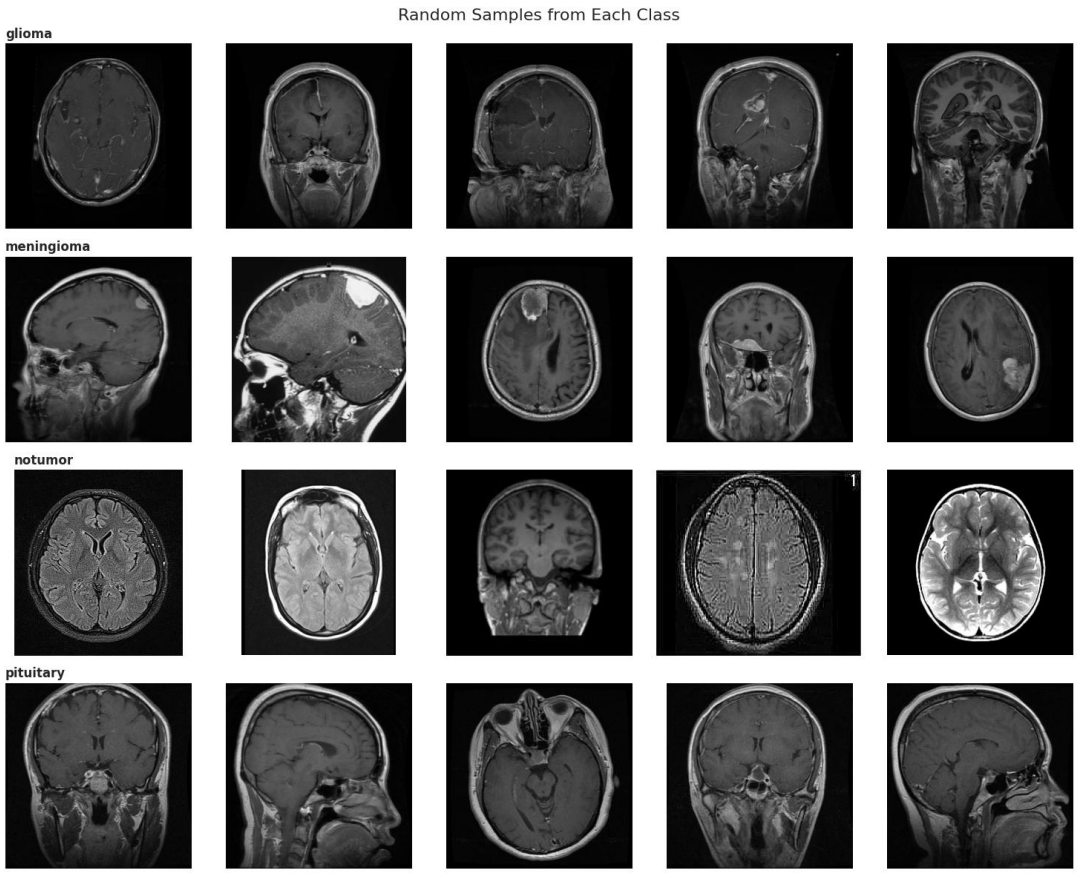


Figure 3. Five random Training images from each class.

E. Pixel Statistics

On the 1,000-image Training sample (grayscale, 128×128, scaled to 0–1), the global mean pixel value is 0.1816 and the standard deviation is 0.1954. The low mean reflects the dark background typical of brain MRI (Figure 4). These two values are used for normalization.

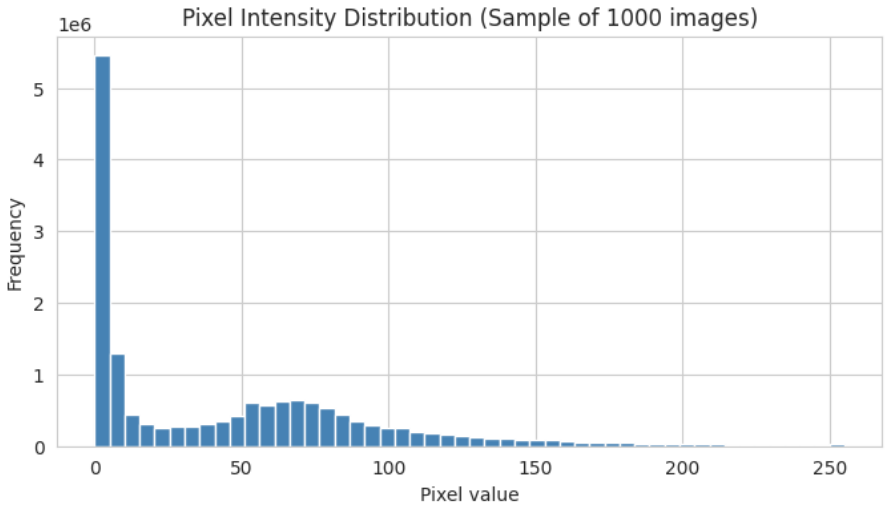


Figure 4. Pixel intensity distribution of a 1,000-image Training sample.

The mean image of each class (200 Training images per class) is shown in Figure 5. In this sample the no-tumor mean image looks more like an axial slice and appears brighter, while the three tumor-class mean images look more like a coronal view. This is only a visual impression from a sample,

but it suggests that class and view may be partly related. If so, a classifier could learn the view or contrast instead of the tumor, which should be checked later (for example with Grad-CAM).

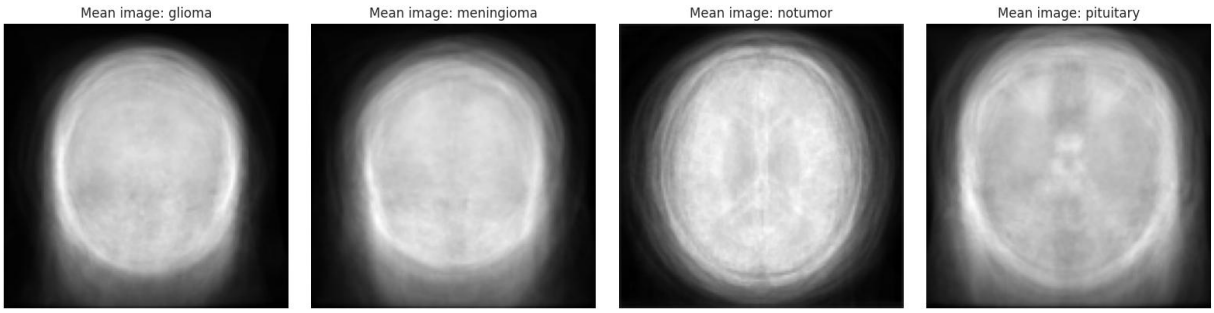


Figure 5. Mean image of each class (200 Training images per class).

F. Data Quality and Cleaning

No corrupted image was found using PIL verification. Exact duplicates were detected using the MD5 hash of the file bytes: 340 images belong to duplicate groups, which corresponds to 187 removable copies (171 inside Training and 16 inside Testing). No hash is shared between Training and Testing, so there is no exact-copy leakage between the splits.

For blur, the variance of the Laplacian was computed on 1,500 sampled images. The distribution is strongly right-skewed (Figure 6): mean 984.99, median 68.99, 25th percentile 50.02, 75th percentile 187.15, maximum 48,579.18 and minimum 5.69. Because the mean is far above the median, the median should be used when choosing any blur threshold. Mean brightness (0–255 scale) has mean 47.33, standard deviation 18.16, minimum 14.94 and maximum 126.73, confirming that most images are dark.

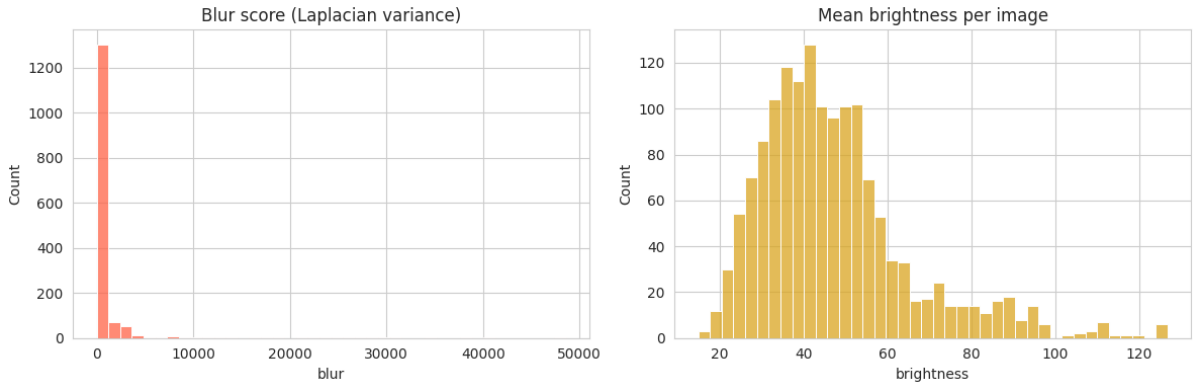


Figure 6. Blur score (Laplacian variance) and mean brightness of a 1,500-image sample.

The five blurriest images in the sample (Figure 7) all still show a clearly visible tumor or brain structure. A low Laplacian variance therefore seems to reflect smooth image content rather than unusable quality, so images should not be removed by a blur threshold alone. Wrong labels cannot be detected automatically and would need visual or expert checking.

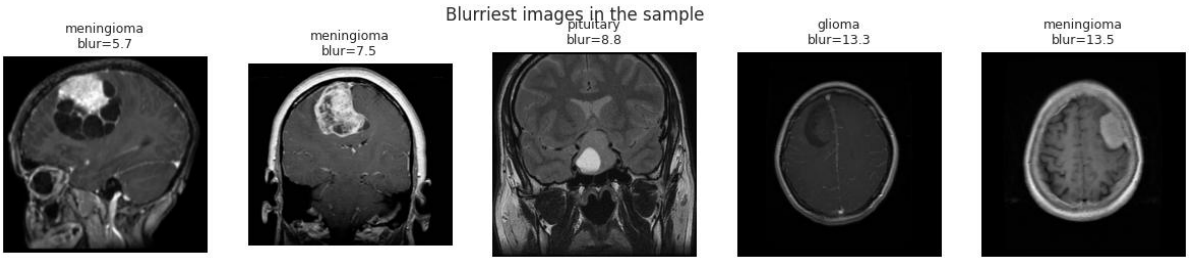


Figure 7. The five blurriest images in the sample (lowest Laplacian variance).

The cleaning rule adopted in the notebook removes duplicates from the Training set only, also removes any Training image that matches a Testing image (none were found), and keeps the Testing set untouched so it remains a fixed benchmark. The result is shown in Table 1 and Figure 8.

Class (Training)	Original	After cleaning	Removed	Class weight
glioma	1,400	1,400	0	0.969
meningioma	1,400	1,386	14	0.979
pituitary	1,400	1,362	38	0.997
notumor	1,400	1,281	119	1.060
Total	5,600	5,429	171	-

Table 1. Training set before and after exact-duplicate removal.

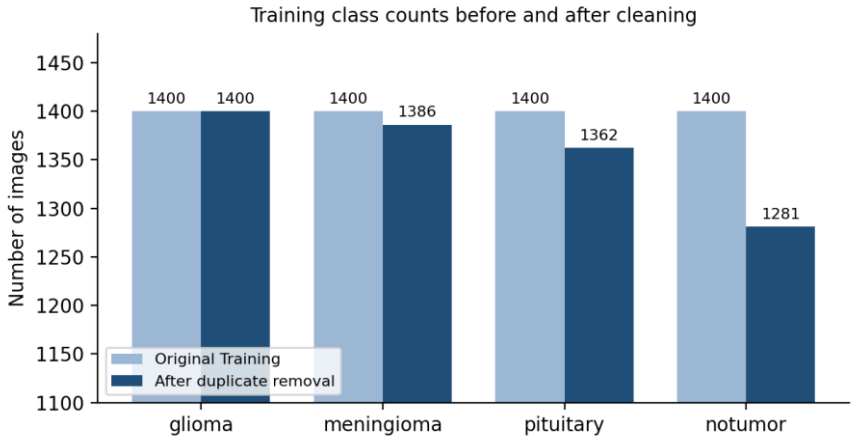


Figure 8. Training class counts before and after duplicate removal.

The no-tumor class lost the most images (119, or 8.5% of its Training images), so the cleaned data is no longer perfectly balanced (max/min ratio 1400/1281 ≈ 1.09). This is mild, and the recomputed inverse-frequency class weights (0.969 to 1.060) can be used if needed. The final dataset has 7,029 images (5,429 Training and 1,600 Testing). The 16 duplicates inside the Testing set were kept so that the test set stays unchanged; this should be mentioned when reporting results.

Limitation: MD5 only finds byte-identical files. Re-saved, resized or cropped copies, and different slices from the same patient, are not detected. The dataset folders contain no patient identifier, so a patient-level split cannot be verified from the data alone.

3. Implications for Modelling and Evaluation

The EDA leads to the following preprocessing and evaluation plan for later tasks:

- **Input size:** resize all images to 224×224 (the most common original size is 512×512, so most images are square).
- **Channels:** convert everything to one format: 3 channels for ImageNet-pretrained models, or 1 channel for a custom CNN.
- **Normalization:** use mean 0.1816 and standard deviation 0.1954 (grayscale, 0–1 scale), computed from Training images only.
- **Cleaning:** use the cleaned Training set (5,429 images) and keep the Testing set (1,600) fixed as the final test set. Do not remove images by blur score alone.
- **Validation:** create a validation split from the cleaned Training set, stratified by class, and never tune on the Testing set.
- **Class weights / augmentation:** the imbalance is small, so weights are optional; augmentation (small rotation, flip, brightness change) is still useful because of the variation in view, contrast and source.
- **Shortcut check:** because class and view/contrast may be related (Section 2E), inspect what the model uses, for example with Grad-CAM.
- **Metrics:** report accuracy together with macro precision, recall and F1, per-class recall and a confusion matrix, since a medical task cares about missed tumors in each class.
- **Fair comparison:** re-run baseline models on our own cleaned split instead of copying numbers from papers.

4. Related Work: Five Brain Tumor MRI Studies

Five IEEE conference papers (2022 to 2026) on CNN-based brain tumor detection from MRI were reviewed. The table below summarizes each paper. The reported values and the checks marked "our check" are taken from our literature comparison table; they were derived from the figures and tables printed in the papers and are not independent re-runs of the models.

Study	Dataset and task	Method and reported result	Main limitations
Sivamurugan et al. (2026) ICISCoIS	Kaggle brain MRI, about 1,300 images stated; 4 classes; 80/10/10 split.	CBAM-Hybrid Net: ResNet50V2, DenseNet121, MobileNetV2 and EfficientNet features with CBAM attention, fused; Grad-CAM and Flask app. Reported accuracy 97.8%, F1 0.98; single-model baselines 93.6% to 95.1%.	Test set of 394 images does not fit a 1,300-image dataset; the confusion matrix gives 380/394 = 96.4%, not 97.8%; fusion method unspecified; no ablation or cross-validation.
Singh et al. (2025) ICDT	Online data plus "real-time" data, not named; 4 classes; stratified 70/30 split.	Transfer learning with VGG16, VGG19, ResNet50 and DenseNet169 (256×256, CLAHE). DenseNet169 best: accuracy 97.45%; ResNet50 92.34%; VGG19 90.23%; VGG16 88.65%.	Dataset unnamed; counts imply about 96.4% for DenseNet169; accuracy and F1 identical for every model; no separate validation set although early stopping is used (leakage risk).

Study	Dataset and task	Method and reported result	Main limitations
Keerthan et al. (2025) SENNET	BraTS and Kaggle MRI (T1, T2, FLAIR); size and split not stated.	Two stages: U-Net segmentation, then GLCM/HOG/LBP features with PCA/LDA and an EfficientNet-based classifier. Segmentation IoU 0.7917 (train) vs 0.3723 (validation); classification metrics printed as "XX%".	Final classification metrics missing; large IoU gap shows weak generalization; curves cover only 3 epochs; inconsistent class definitions.
Anshul et al. (2024) ICAC2N	Dataset not named; binary tumor / non-tumor; 80/20 split.	Skull masking, wavelet and GLCM features, segmentation, then a custom CNN with sigmoid output. Accuracy 95%; cited baselines SVM 85% and DNN 88%.	No sample counts; baselines come from other datasets (not like-for-like); no recall, confusion matrix, cross-validation or ablation.
Yadav et al. (2022) CISES	Kaggle MRI, 1,500 tumor + 1,500 healthy (also described as 3,060 images); 64×64; binary.	Transfer learning with Inception-V3, ResNet50 and VGG-16. VGG-16 best: accuracy 0.96, F1 0.97; ResNet50 0.89; Inception-V3 0.75; training time also reported.	Contradictory dataset description; low 64×64 resolution; ROC-AUC equals accuracy for all models; training curves stay below the reported 0.96; split not given.

Table 2. Summary of the five reviewed studies.

Reported results at a glance

Paper (year)	Task	Best model	Reported	Our check
Sivamurugan (2026)	4-class	CBAM-Hybrid Net	97.8%	96.4% from confusion matrix (380/394); test size inconsistent with dataset size
Singh (2025)	4-class	DenseNet169	97.45%	About 96.4% from per-class counts; accuracy and F1 identical
Keerthan (2025)	Segmentation + classification	U-Net + EfficientNet-based CNN	Not reported ("XX%")	IoU 0.79 (train) vs 0.37 (validation)
Anshul (2024)	Binary	Custom CNN + segmentation	95%	Not verifiable; baselines taken from other datasets
Yadav (2022)	Binary, 64×64	VGG-16	96%	Table consistent; training curves stay below the reported value

Table 3. Reported accuracy compared with our consistency check.

These scores are contextual only. The papers use different datasets, class definitions and splits, so they cannot be compared directly with each other or with our future results. The ranking of architectures also changes between papers (for example VGG-16 is best in Yadav et al. but weakest in Singh et al.), which shows that rankings depend on the experimental setup.

5. Research Gap and Proposed Direction

Four recurring problems appear across the five studies:

- Dataset and split reporting is incomplete: dataset names, sample counts, split ratios and hyperparameters are often missing or inconsistent, and some reported numbers do not match the papers' own confusion matrices or counts.
- Evaluation is weak: few studies use a separate validation set, cross-validation, patient-level splitting or external validation, so leakage and optimistic accuracy are possible.

- Few controlled comparisons: baselines are often taken from other papers on different data, and ablations (for example of attention or segmentation) are missing.
- Practical aspects are rarely measured: inference time, model size, explainability and cross-dataset generalization.

Our dataset is public, balanced, four-class and has a fixed test set, which makes a transparent comparison possible. After removing duplicates and confirming zero exact train/test overlap, a proposed follow-on study is to train the same CNN architectures (for example a custom CNN, ResNet50, DenseNet121, EfficientNet) under one identical preprocessing and split, report macro-F1, per-class recall and confusion matrices, and also record training time and inference time. Grad-CAM can be added for explainability. These experiments are proposals for later tasks and were not performed in Task 1.

6. Conclusion

Task 1 produced a reproducible EDA baseline for the Brain Tumor MRI Dataset. The data has 7,200 images in four perfectly balanced classes, no corrupted files and no exact overlap between Training and Testing. The main concerns are the strong variation in image size, color mode, view and contrast (447 sizes, mixed grayscale and RGB, axial/sagittal/coronal views) and the 187 exact duplicates, which were handled by cleaning the Training set only and keeping the Testing set fixed. A 224×224 resize, mean/std normalization from Training data, a stratified validation split and macro-level metrics are justified for the next tasks. The literature review shows that many published results on similar data are hard to verify, so we will re-run all baselines ourselves. No classifier was trained here, so no accuracy claim is made.

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